

NCTU-EE IC LAB – Fall 2015

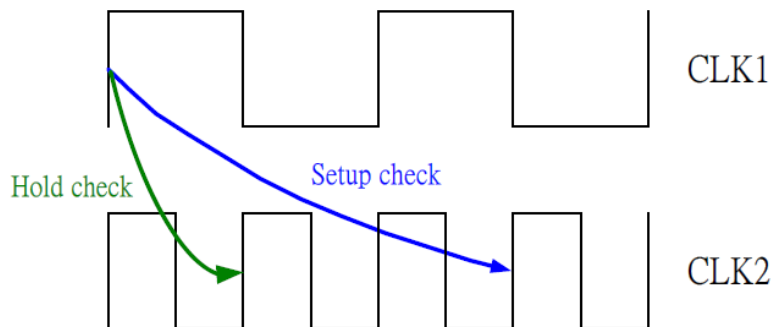
Midterm Online Test - Quiz 1:00p.m. ~ 2:00p.m.

1. [2pts] What is the difference between sequential circuit and combinational circuit from the hardware architecture's point of view?

Ans:

Sequential circuit is used to describe the behavior of flip flops under the clock, and it has memory element which can retain values, while the combinational circuit is consisted by Boolean circuits, and the output is a pure function of input only.

2. [5pts]
- (a) Please explain what setup time is and what hold time is, respectively.
- (b) How can we solve the setup time violation and hold time violation problem happened in the synthesis part?
- (c) If we want to set multicycle path as the following waveform, what value should be filled in the parentheses?
- ```
set_multicycle_path () -from CLK1 -to CLK2
set_multicycle_path () -hold -end -from CLK1 -to CLK2
```



Ans:

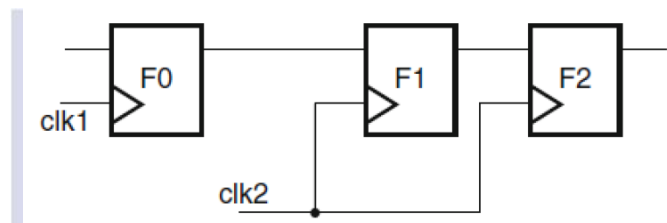
- (a) Setup time is the time that a signal should arrive and become stable at the cell before the clock triggered. Hold time is the time that a signal should retain its value after the clock triggered. (2pts)
- (b) We can modify the RTL code to reduce the delay of the critical path, or increase the clock period to solve the setup time violation. And the hold time violation can be solved by adding buffers on the violated path. (2pts)
- (c) 3, 1. (1pts)

3. [6pts]

- (a) What is CDC (Clock Domain Crossing)?
- (b) What is metastability? Why we may meet this problem in CDC cases?
- (c) Please draw the block diagram of Double Flop Synchronizer, and give an example of using it.

Ans:

- (a) When the data launched and captured by different (asynchronous) clock domain, this case is called CDC. (2pts)
- (b) The unstable status due to non-ideal data transition is called metastability. In the CDC cases, sometimes the input of a clock triggered flip flop will come from another clock domain, and the time between the edges of different clocks may be uncertain, this may cause the metastability. (2pts)
- (c) We can use it as a flag to determine whether the data is stable or not. (2pts)



- 4. [2pts] What is false path? Please give an example that we could set it.

Ans:

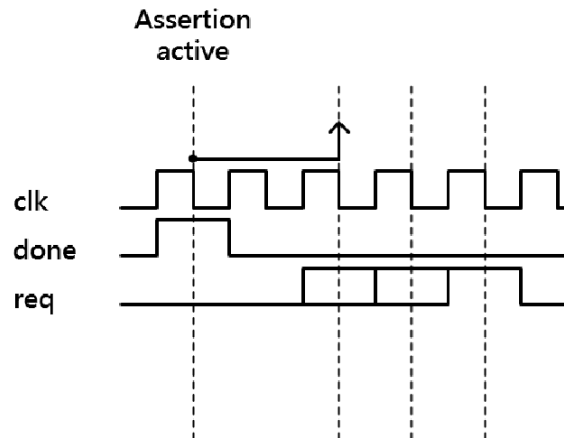
False paths are timing paths that will never really be exercised in the final design. In the common CDC cases, the path between two different clocks is a false path.

- 5. [2pts] Please explain the meaning of the following assertion (better with waveform):

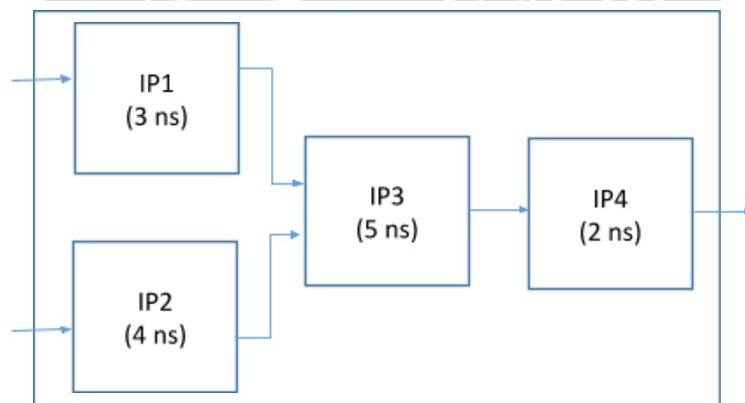
```
property next_req;
 @(negedge clk) done == 1 | => ##[1:3] req == 1;
endproperty
```

Ans:

This assertion will check at negative edge of *clk*. For non-overlapped implication, the consequent sequence expression is evaluated on the next clock tick. So, this assertion checks that whenever *done* is asserted, *req* must be asserted after two cycles, or the following 2 cycles.



6. [2pts] Assume the combinational part of a circuit is shown as the following figure. All the IPs in the figure is all combinational, while the IP1 need 3ns to calculate the result out, IP2 need 4ns, IP3 need 5ns and IP4 need 2ns. If we apply pipeline design concept on this circuit to get smaller clock period:
- How many nanosecond will the minimum clock period be? (Assume there won't be any additional control delay if pipelined)
  - How many cycles will be taken for calculating 5 results out?



Ans:

(a) 5ns. (1pts)

(b) 7 cycles. (1pts)

7. [4pts] IP can be divided as 3 types: soft IP, hard IP and firm IP. Please explain soft IP and hard IP respectively.

Ans:

Hard IP: GDSII format, high performance but technology dependent.

Soft IP: RTL design, requires verification.

8. [4pts] Please explain the advantages and disadvantages of synchronous reset

and asynchronous reset, respectively. (At least one description for each advantage/disadvantage.)

Ans:

Synchronous reset:

%Pros: Not easy to be interfered, easier for simulators and timing analysis.

%Cons: Larger area, not timely.

Asynchronous reset:

%Pros: Timely, smaller area.

%Cons: Easy to be interfered by noise, may cause metastability if the reset signal is near the trigger edge of the clock.

9. [4pts]

(a) What kinds of information are included in the Verilog file generated by the synthesis tool ("DESIGN\_SYN.v") and the sdf file ("DESIGN.sdf") generated by the "write\_sdf" command?

(b) What's the difference between the waveform files generated in the RTL simulation and gate-level simulation?

Ans:

(a) DESIGN\_SYN.v is composed of cells in target library and connection wires. The sdf file contains the correct timing check, IO delay, and wire delays of each instance in DESIGN\_SYN.v. (2pts)

(b) The waveform files generated in RTL simulation only contain the correctness of the logic in the design file. The waveform files in gate-level simulation contain the output results and delays of instances in the netlist file, so we can see the actual delay of signals in it. (2pts)

10. [4pts] Assume we have a 7-bit signal "temp" and a 3-bit signal "volt" in the interface of a design. Now we want to set three types of the value of temp: 0~33 is low, 34~66 is medium, and 67~127 is high. And we want to set the cross bin of temp and volt. Please complete the following code:

```
covergroup cov;
 temperature: coverpoint inf.temp {
 bins low = "1.write your code here";
 bins medium = "2.write your code here";
 bins high = "3.write your code here";
 }
 voltage: coverpoint inf.volt;
 cross_bin_check: "4.write your code here";
```

endgroup

Ans:

1. {[0:33]}
2. {[34:66]}
3. {[67:127]}
4. cross temperature, voltage;

